Mining Temporal Data: A Coal-Fired Boiler Case Study

Paper Summary

SUMMARY PASSAGE 1

Introduction

The ability to predict and avoid rare events in time series data is a challenge that could be addressed by data mining approaches. Difficulties arise from the fact that often a significant volume of data describes normal conditions and only a small amount of data may be available for rare events. This problem is further exacerbated by the fact that traditional data mining does not account for the time dependency of the temporal data.

SUMMARY PASSAGE 2

Step 1: Parameter Categorization

A time widow is defined as a set of observations in chronological order that describe a specified amount of continuous observations. This step allows the data mining algorithms to account for the temporal nature of the data. The most effective method to segment the data is by determining/estimating the approximate date of failure and set that as the last observation of the final time window.

SUMMARY PASSAGE 3

Step 3: Statistical And Visual Analysis

This step involves statistical analysis of the data in each time period that was designated in the previous step. Process shifts, changes in variation, and mean shifts in parameters are helpful in indicating that the appropriate time windows and parameters were selected.

SUMMARY PASSAGE 4

Step 4: Knowledge Extraction

Data mining algorithms discover relationships among parameters and an outcome in the form of IF â€¦ THEN rules and other constructs (e.g., decision tables) [1], [5]. Data mining is natural extension of more traditional tools such as neural networks, multivariable algorithms, or traditional statistics. In the detection of rare events, the decision-tree and rule-induction algorithms are explored for two significant reasons.

SUMMARY PASSAGE 5

Step 5: Analysis Of Knowledge And Validation

This step deals with validation of the knowledge generated by the data mining algorithm. If a validation data set is available it should be used to validate the accuracy of the rules. If no similar data is available then unused data from the analysis or a 10-fold crossvalidation can be utilized [6].

SUMMARY PASSAGE 6

Power Boiler Case Study

The rules accurately predicted the normal cases, but they were not as effective in predicting the fault cases. This is most likely explained by the fact that the test data labeled, time window 6, was extracted after the boiler had been derated. The derating of the boiler significantly changes the combustion process and was not included in the original data set.

SUMMARY PASSAGE 7

Future Research

The approach presented in this research produced rule sets that can be utilized for the development of a meta-control system. Integrating concepts from expert advisory systems and intelligent power control systems will form the meta-control system architecture for the avoidance of the ash pluggage.

SUMMARY PASSAGE 8

Conclusion

This approach was applied to a commercial tangentially-fired coal-boiler to detect and avoid an ash fouling pluggage that eventually leads to boiler shutdown. The approach produced a rule set that was over 99.7% accurate. The knowledge base was also validated with a separate test data set that has predicted failures with accuracy of over 89.8%.